

① check:

one  $\$a0$ ,  $\$a1$ , else-case

if-case:

li  $\$t0$ , 1

j exit-check

else-case:

li  $\$t0$ , 0

exit check:

jr,  $\$ra$

②

li  $\$s0$ , 0

li  $\$t0$ , 0

Loop:

lge  $\$t0$ ,  $\$s1$ , EndLoop

slr  $\$t2$ ,  $t0$ , 2

add  $\$t2$ ,  $\$t2$ ,  $\$s2$

lw  $\$t3$ , 0( $\$t2$ )

add  $\$s0$ ,  $\$s0$ ,  $\$t3$

addi  $\$t0$ ,  $\$t0$ , 1

j Loop

EndLoop:

④  $y = n - A[i];$

⑤ addi  $\$t0$ ,  $\$s1$ , -2

slr  $\$t0$ ,  $\$t0$ , 2

add  $\$t0$ ,  $\$t0$ ,  $\$s3$

lw  $\$t1$ , 0( $\$t0$ )

addi  $\$t1$ ,  $\$t1$ , 5

slr,  $\$t2$ ,  $\$s0$ , 2

add \$t2, \$t2, \$v2  
sw \$t1, 0(\$t2)

⑥

i = \$t1  
result = \$v2  
memarray = \$s0

```

1      li    $t1, 0    i = 0
2  LOOP:
3      lw    $s1, 0($s0)  S1 = memarray
4      add   $s2, $s2, $s1  result += v1
5      addi  $s0, $s0, 4  # increments elements
6      addi  $t1, $t1, 1  i++
7      slti  $t2, $t1, 100  t2 = (i < 100)
8      bne   $t2, $zero, LOOP  t2 = i < 100

```

int i;

```

for (i = 0; i < 100; i++) {
    result += memarray[i];
}

```

⑦

i = \$v1  
n = \$v2  
Values = \$s0

```

move $s1, $s2  i = n
LOOP:
    blez $s1, END_LOOP  i <= 0
    sll  $t0, $s1, 2  t0 = i * 4
    add  $t1, $s0, $t0  t1 = Values[i]
    lw   $t2, 0($t1)  t2 = Values[i]
    sll  $t2, $t2, 1  t2 = t2 * 2
    sw   $t2, 0($t1)  Values[i] = t2
    addi $s1, $s1, -1  i--
    j    LOOP
END_LOOP:

```

```

for (i = n; i > 0; i--) {
    Values[i] = Values[i] * 2;
}

```

⑧

$D[i] = A[i] + B[i]$ ;

i = \$s0  
j = \$s1  
A = \$s2  
B = \$s3  
C = \$s4

```

    rli $t0, $s0, 2
    add $t0, $t0, $s2
    sw  $t0, 0($t0)
    rli $t2, $s1, 2
    add $t2, $t2, $s3
    rli $t3, 0($t2)

```

```

    add $t4, $t1, $t3
    rli $t0, $s0, 2
    add $t0, $t0, $s4
    sw  $t4, 0($t0)

```

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$g = \$r0$   
 $n = \$r1$   
 $i = \$r2$   
 $A = \$r6$

```
addi $t0, $s2, 4  $t0 = i + 4$ 
sll  $t0, $t0, 2  $t0 = (i + 4) * 4$ 
add  $t0, $s6, $t0  $t0 = A[i + 4]$ 
lw   $t1, 0($t0)  $t1 = A[i + 4]$ 
sub  $s0, $t1, $s1  $g = A[i + 4] - n$ 
```

$g = A[i + 4] - n;$

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$i = \$r1$   
 $A = \$r0$

```
addi $s1, $zero, -1  $i = -1$ 
LOOP:
addi $s1, $s1, 1  $i++$ 
sll  $t0, $s1, 2  $t0 = i * 2$ 
add  $t1, $s0, $t0  $t1 =$ 
lw   $t2, 0($t1)  $t2 = A[i]$ 
beq  $t2, $zero, LOOP  $A[i] == 0$ 
```

$i = -1;$

do {

$i++;$

} while ( $A[i] == 0$ );

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$i = \$r0$   
 $n = \$r1$   
 $A = \$r2$   
 $B = \$r3$

for ( $i = 0; i < n; i++$ ) {  
 $B[i] = A[n - 1 - i];$   
}

li \$r0, 0  $\# i = 0$

Loop:

bge \$r0, \$r1, endloop

sub \$t0, \$r1, \$r0

addi \$t0, \$t0, -1

sw \$t0, \$r3, 2

add \$t1, \$t1, \$r2

sw \$t2, 0(\$t1)

sl \$t3, \$r0, 2

add \$t3, \$t3, \$r3

sw \$t2, 0(\$t3)

addi \$r0, \$r0, 1

j loop

endLoop: